

Literature-Implied Resolution of Three Problems on Hypercyclic Matrix Tuples

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11 August 2026

Status: literature-implied answers to all three problems. For upper triangular Toeplitz matrices the sharp minima are $2n$ over $\mathbb{C}$ and $n + 1$ over $\mathbb{R}$. For $n \geq 3$, no commuting family of real $n \times n$ matrices can have every nonzero vector hypercyclic.

1 Source questions

Costakis, Hadjiloucas, and Manoussos, *Dynamics of tuples of complex upper triangular Toeplitz matrices*, arXiv:1008.0780v2 [1], ask on arXiv PDF page 11 for the minimal number of such matrices over $\mathbb{C}$ and $\mathbb{R}$ (Problems 1–2), and whether for $n \geq 3$ there is a commuting real n -tuple for which every nonzero vector is hypercyclic (Problem 3).

$$\left\{ \left(\begin{array}{c} \sum_{j=1}^4 k_j \frac{a_{j,2}}{a_{1,1}^{k_1} a_{2,1}^{k_2} a_{3,1}^{k_3} a_{4,1}^{k_4}} \\ \sum_{j=1}^4 k_j \frac{a_{j,2}}{a_{1,1}^{k_1} a_{2,1}^{k_2} a_{3,1}^{k_3} a_{4,1}^{k_4}} \end{array} \right) : k_1, k_2, k_3, k_4 \in \mathbb{N} \right\}$$

is dense in $\mathbb{C}^4$.

Proposition 6.3. *The tuple (A_1, A_2, A_3, A_4) is not hypercyclic.*

7. OPEN PROBLEMS

Problem 1. What is the minimal number of complex upper triangular Toeplitz matrices that form a hypercyclic tuple in $\mathbb{C}^n$? Observe that in [6] the authors have shown the existence of an eight tuple of complex upper triangular 2×2 Toeplitz matrices being hypercyclic in $\mathbb{C}^2$, which is far from being optimal.

Problem 2. What is the minimal number of real upper triangular Toeplitz matrices that form a hypercyclic tuple in $\mathbb{R}^n$? In [7] the authors have shown that the minimal number for 2×2 matrices is 3.

Problem 3. In [6] an example of a two-tuple (A_1, A_2) of commuting 2×2 matrices is given in which every non-zero vector in $\mathbb{R}^2$ is hypercyclic for (A_1, A_2) . In other words we say that the $\mathbb{N}^2$ action is minimal. Does the above extend in any dimension $n \geq 3$? That is, for $n \geq 3$, does there exist a hypercyclic n -tuple $(A_1, \dots, A_n)$ of commuting $n \times n$ matrices for which every non-zero vector x in $\mathbb{R}^n$ is hypercyclic?

Figure 1: The three questions on source PDF page 11.

2 Problems 1–2: exact Toeplitz counts

Abels and Manoussos, *Topological generators of abelian Lie groups and hypercyclic finitely generated abelian semigroups of matrices*, arXiv:1108.1123 [2], compute in Table 3 on arXiv PDF page 11 the minimum number of generators for the groups of invertible $n \times n$ Toeplitz matrices:

$$\boxed{2n \text{ over } \mathbb{C}}, \quad \boxed{n + 1 \text{ over } \mathbb{R}}.$$

Their proof identifies the invertible Toeplitz group as an abelian Lie group with a dense orbit and applies their topological-generator theorem. The complex group has real noncompact dimension $2n - 1$ (its maximal torus is S^1), while the identity component of the real group has noncompact dimension n ; a connected abelian Lie group of noncompact dimension d needs and admits $d + 1$ dense semigroup generators.

There is a small identification needed because the source problems do not require the generators to be invertible. Let J be the nilpotent backward shift and write a triangular Toeplitz matrix as

$$A = a_0 I + a_1 J + \cdots + a_{n-1} J^{n-1}.$$

Its last coordinate acts by multiplication by a_0 , so A is singular exactly when $a_0 = 0$. If a product uses a singular generator, its image has last coordinate zero. Consequently, inside the open set

$$U = \{x \in \mathbb{K}^n : x_n \neq 0\}, \quad \mathbb{K} \in \{\mathbb{R}, \mathbb{C}\},$$

every point of a tuple orbit comes from the subtuple of invertible generators. If the full orbit is dense, that invertible subtuple already has a dense orbit in U . Conversely, the invertible Toeplitz group acts simply transitively on U via $A \mapsto Ae_n$, and U is dense in $\mathbb{K}^n$. Thus singular generators cannot lower the minimum, and the two entries of Abels–Manoussos Table 3 answer Problems 1–2 exactly.

3 Problem 3: a common-eigenline obstruction

The negative answer follows from the classical simultaneous-eigenvector argument for commuting complex matrices.

Proposition 1. *Let $n \geq 3$ and let $A_1, \dots, A_r \in M_n(\mathbb{R})$ commute. There is a nonzero proper real subspace $W \subset \mathbb{R}^n$, of dimension at most two, invariant under every A_j . Hence no such family can have every nonzero vector hypercyclic. In particular, the answer to Problem 3 is no.*

Proof. Complexify the matrices. A finite commuting family of complex matrices has a common eigenvector: choose an eigenspace of A_1 ; commutativity makes it invariant under all remaining matrices, then successively choose eigenspaces of their restrictions. Hence there are $v \in \mathbb{C}^n \setminus \{0\}$ and $\lambda_j \in \mathbb{C}$ such that

$$A_j v = \lambda_j v \quad (1 \leq j \leq r).$$

Put

$$W = \text{span}_{\mathbb{R}}\{\text{Re } v, \text{Im } v\} \subseteq \mathbb{R}^n.$$

This is nonzero and has dimension at most two. Since every A_j is real, the displayed eigenvalue equations show that $A_j W \subseteq W$. When $n \geq 3$, W is proper. The orbit of any $x \in W \setminus \{0\}$ under the generated semigroup remains in the closed proper subspace W , so it cannot be dense in $\mathbb{R}^n$. $\square$

4 Provenance, checks, and scope

Problems 1–2 are agent-identified consequences of the later Abels–Manoussos theorem; that paper does not cite arXiv:1008.0780, so this is classified as *literature implied*, not an author-explicit literature answer. Problem 3 is an immediate application of the standard common eigenvector theorem, with the complete deduction included above. No new mathematical result is claimed by this packet.

The local registry, solution, attempt, and proof-gap indexes were searched by both arXiv ids, titles, and the phrases “upper triangular Toeplitz”, “minimal number”, and “every non-zero vector hypercyclic”. Targeted arXiv searches additionally found the general finite-dimensional tuple papers of Shkarin and Ayadi–Marzougui, but no contradictory count or positive dimension- $n \geq 3$ minimal action. The key human-review checks are the invertible-subtuple reduction and the interpretation of Table 3.

References

- [1] G. Costakis, D. Hadjiloucas, and A. Manoussos, *Dynamics of tuples of complex upper triangular Toeplitz matrices*, arXiv:1008.0780v2 (2010).
- [2] H. Abels and A. Manoussos, *Topological generators of abelian Lie groups and hypercyclic finitely generated abelian semigroups of matrices*, Adv. Math. **229** (2012), 1862–1872; arXiv:1108.1123.